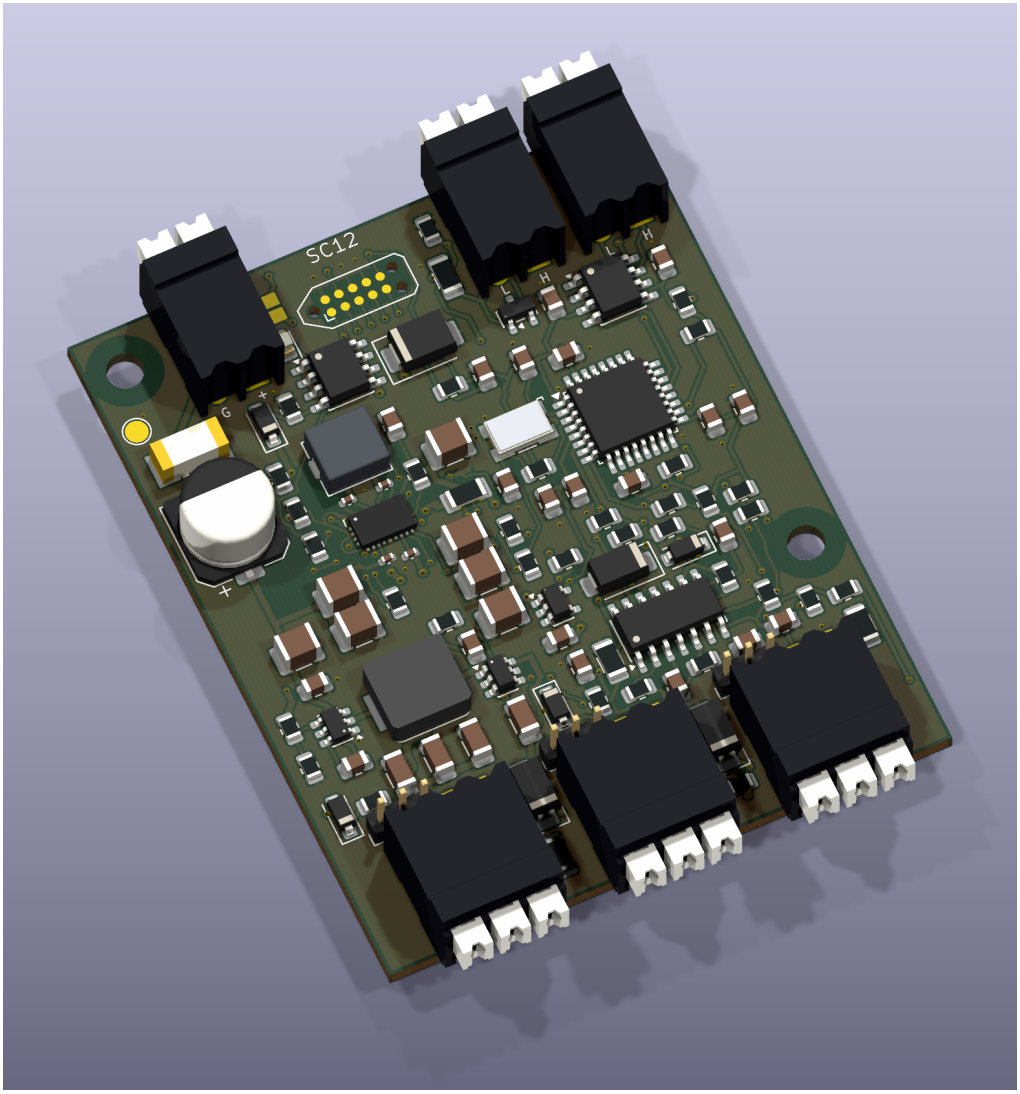
FRC
Sensor CAN

Revision D-2 SC12 · Prototype engineering and assembly guide

Three industrial / logic sensor inputs
Regulated 12 V and 5 V sensor power



2026-09-08

Editable KiCad files, split factory/hand-solder BOMs, firmware, robot examples and verification records accompany this guide. No physical prototype has been tested. Electrical ratings are design targets pending the included qualification plan.

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1 Product overview

Source: README.md

Rev D-2 preserves Rev D's circuit and normal telemetry identity (hardware 4, protocol 3). It changes 27 passive placements to specified equivalents, adds three removable power-selection headers, and separates factory SMT assembly from your through-hole work. Cost basis and remaining costs.

Three industrial digital inputs, selectable **regulated 12 V or 5 V sensor power**, and classic CAN for FRC team projects. The **52 × 64 mm** PCB keeps horizontal Weidmüller push terminals and the compact TC2050 programming contacts. **Every component mounts on the top side.** It uses four copper layers, with both inner layers reserved for ground.

Engineering prototype release. CAD, source, manufacturing files and checks are provided for review and prototype fabrication. No physical Rev D-2 board has been assembled or qualified. Production release and competition use require the tests in VALIDATION.md; CAD checks do not establish either.

Item	Rev D-2 design target
Board input	6-22 V DC at J1, protected distribution branch required
Sensor supplies	12.01 V nominal buck-boost and 5 V nominal; 300 mA shared per rail
Inputs	Three PNP, NPN or push-pull inputs; positive signal range 0-30 V
Logic thresholds	LOW ≤1.2 V, HIGH ≥2.4 V targets; supports 3.3/5 V logic, not guaranteed 1.8 V
Connectors	Weidmüller LSF-SMT 3.50/90; 1824420000 and 1824430000
MCU / CAN	STM32F042K6T6 LQFP32 + TCAN3413 SO8; 1 Mbit/s
Power stage	TPS552872-Q1, synchronous buck-boost; 3.3 µH shielded inductor
Assembly	107 fitted parts: 98 top-side factory SMT; six terminals and three headers hand soldered; three loose caps
Parts estimate	$40.40 / $34.69 / $32.23 per board in lots of 5 / 10 / 25, including purchase minimums; PCB/assembly/freight/tax extra
Firmware	0.5.0 + bootloader 1.0.0; hardware 4/protocol-v3 telemetry; CAN setup/update CLI and GUI

12 V remains regulated when the battery is above or below 12 V within the qualified input/load envelope. It is intended for compatible three-wire Omron/Banner sensors; verify the exact model, current, output type and residual/leakage voltage. A brand name alone is not a compatibility guarantee. Sensor power is hardware-enabled; removable 2 mm jumper caps choose each port's supply. No raw battery output is exposed on Rev D-2.

Firmware reports qualified 12 V power-good and 5 V switch fault status. Counts pause on either rail fault and resume only after 100 ms of continuous good indications. Drivers provide a health-checked input helper. These indications are not calibrated voltage measurements and cannot detect every cable or sensor failure.

The integrated switches, current limiting, shared 5 V driver bias and PCB Kelvin taps avoid extra purchased power devices. Most passives remain 0805 or larger. Four local 0402 capacitors and the 21-pad HotRod QFN require a stencil and controlled reflow/heatbed; inspect the power IC joints. Hand soldering the remaining parts is practical. See design choices.

Order and finish a board

1. Request top-side SMT assembly using the Gerber ZIP, JLCPCB BOM and CPL. Match all 98 placements against Factory_BOM.csv; a blank factory part ID still needs exact-MPN sourcing. Follow the factory instructions.

2. Buy and hand solder the six terminals and three headers, and buy three closed-top caps. The factory leaves these nine through-hole positions empty. A complete placement drawing also shows these user-fitted parts, so use the SMT-only CPL for the factory order.

3. Leave the caps off during initial bring-up. Follow assembly and programming, then configure voltage, bias and termination with power disconnected. Use the firmware and drivers supplied here; D-2 retains D's hardware-4/protocol-v3 identity.

Board quantity	Parts purchase cost per board
1	$55.27
5	$40.40
10	$34.69
25	$32.23

These estimates include all fitted parts, loose caps and supplier minimums, using observations from September 7-8, 2026. PCB, assembly, attrition, freight and tax are extra; **a delivered price below $50 has not been established**. The $44.16 first-tier consumed-parts figure is a different basis from purchasing enough parts for one board. Calculation and assumptions.

Confirm component allocation and a destination-specific express-delivery estimate against the one-month target. The factory guide treats 2-4 weeks as a planning allowance, subject to sourcing, production and customs; no order or delivery date is confirmed.

Removable sensor-voltage selection

JP101, JP201 and JP301 select power for J4, J5 and J6 respectively. Fit **one 2 mm cap per header** with input power disconnected:

Cap position	Sensor power
Pins 1-2	Regulated 5 V
Pins 2-3	Regulated 12 V
Cap removed	Port power disconnected

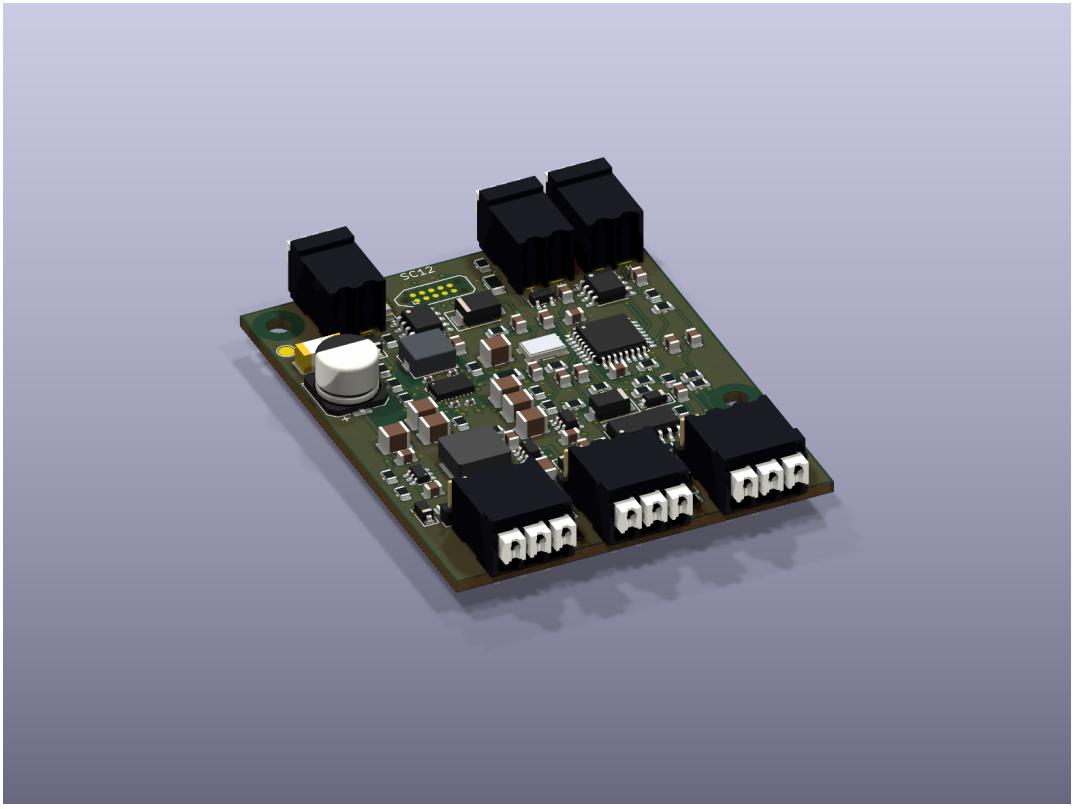
Pins run 1-2-3 left to right when viewed from the component side; pin 1 has a square PCB pad. Bias selectors JP103/JP203/JP303 and CAN termination JP2 remain solder settings. The caps are friction-retained, with retention/vibration qualification still required for the actual robot installation. Full wiring and retention instructions.

Files and verification

The SC12 optimization plan records implementation and open qualification gates. CAN servicing now provides full-UID discovery, blink, persistent ID/settings and recoverable firmware uploads. Use the CLI or desktop window through SocketCAN or the robot bridge. Initial SWD installation is required. Software tests pass; real-board/controller, power-cut and vibration qualification remain pending.

Deliverable	Link
Native KiCad PCB / schematic	PCB, schematic
Manufacturing / assembly	Gerbers and drills, interactive BOM, schematic PDF
Ordering	Gerber ZIP, SMT BOM, JLCPCB BOM, CPL
User-fitted parts	Hand-solder BOM, loose caps, DigiKey import
Prototype factory packet	ZIP
Complete source/review release	ZIP, hashes
Mechanical handoff	STEP, dimensioned drawing

Deliverable	Link
Setup and operation	wiring, assembly, protocol, drivers
Engineering checks	Consolidated report, D-to-D-2 comparison, SMT file checks, qualification plan
Printable handbook	PDF



Connector geometry and official model provenance. Case design is a subsequent task after review of this revision; the mechanical files are a nominal handoff, not an enclosure fit test. Earlier A/B/B-2/C/D designs remain available in the repository.

2 Simplification

Source: docs/SIMPLIFICATION.md

Rev D-2 retains Rev D’s regulated 12 V while retaining three inputs and a 52 × 64 mm PCB. Compared with Rev C's 52 × 56 mm, area grows 14.3%; it is still 18.4% smaller than the 60 × 68 mm B versions. Rev C has 89 fitted parts; D has 104 and D-2 has 107 plus three loose caps. The increase buys regulation across the battery's 12 V crossing and explicit power-health handling.

What saves space and parts

• A four-switch synchronous TPS552872-Q1 integrates the buck-boost switches and output-current-limit control. It replaces the raw-battery rail and its separate TPS1H200A switch, current-set resistor and three capacitors.

• Its driver uses the existing internal 5 V buck rail, avoiding the high-input-voltage loss of its internal linear bias regulator. The 5 V design reserves 100 mA for this supply. Startup and brownout sequencing must be checked on the prototype.

• One 0.1 ohm 1206 resistor sets the 12 V current limit. Separate PCB Kelvin taps carry the feedback signals; they are copper features, not extra components. This resistor dissipates about 9 mW at 300 mA.

• Both internal copper layers are ground. Signal and power routing stay on the outer layers. The return planes and local ground vias help a compact switching layout; this is not a measured EMI claim.

• MCU internal pullups read the two open-drain health outputs without two external resistors. The converter's unused current-limit and cable-compensation status/functions are omitted.

• 10k compensation resistance reuses an existing BOM line. Firmware inversion/debounce, fixed input thresholds and one status LED retain Rev C's simpler interface.

Why the remaining capacitors matter

The converter has two bootstrap capacitors, two high-frequency supply bypasses, input/output energy-storage banks, a driver-bias capacitor and compensation capacitor. They serve different current paths. Moving or merging a bootstrap/bypass capacitor into a distant bulk capacitor is not equivalent.

The six added 10 µF 50 V X7R 1210 capacitors share one MPN. Samsung's typical bias curves plus explicit tolerance, temperature and aging allowances give approximately 7.10 µF for the two-cap input bank at 22 V, 15.32 µF for the three-cap output bank at 12.22 V, and 6.20 µF for driver bypass at 5.22 V. TI calls for at least 4.7 µF effective input, 10 µF output and 4.7 µF bias capacitance. These are engineering budgets from typical curves, not manufacturer-guaranteed minima. Substitution needs the same review.

The existing three 22 µF capacitors on the 5 V buck retain its effective-capacitance budget. Input TVS/filter/feedback parts handle industrial wiring and noise. Each MCU/CAN/comparator bypass remains at its local supply pins. Numerical checks record the assumptions.

Assembly compromise

There are 39 resistors, 37 capacitors and 31 other fitted parts. Only four capacitors are 0402; other small passives are largely 0805, with 1206/1210 parts where power or effective capacitance requires them. The converter is a 5 × 3 mm HotRod QFN with five long electrical power pads, not a generic exposed-pad QFN. Its footprint follows TI drawing RYQ0021B. A laser stencil, controlled reflow and joint inspection are required for repeatable assembly. This is less forgiving than Rev C's mostly leaded packages.

No per-port electronic power switch, output driver, encoder interface, ADC voltage/current telemetry, wireless radio or USB connector is added. All three ports share each rail's load budget. There are three removable 2 mm power headers, three bias solder selectors and one optional CAN termination bridge. The user hand solders all nine through-hole parts; the factory places 98 SMT parts. Procurement changes and cost basis.

Efficiency and stability scope

Nominal switching frequency is 781.25 kHz; automatic light-load mode is enabled. The shielded 3.3 µH inductor is sized for converter peak-current protection, not just the 300 mA load rating. The low-value shunt and external 5 V gate bias reduce avoidable loss. No measured efficiency percentage is claimed.

The boost continuous-conduction analytical model screens 192 corners using TI equations 19-26, capacitor derating, inductance tolerance, a ±20% transconductance allowance and an assumed full-cycle delay. It is not a switching simulation. PFM, current limiting, line steps, load steps, startup and the buck/boost transition still require measurement. See loop report and plot.

At these modest sensor loads, operation can enter discontinuous/light-load mode; the 192-corner CCM model is a screening model rather than a prediction of the normal light-load waveform. The 3.3 µH choice keeps the inductor small with adequate saturation margin, while automatic light-load operation avoids unnecessary circulating current. Measure ripple and input power with the actual sensors.

3 Wiring and programming contacts

Source: docs/WIRING.md

Use the included hardware-4/protocol-v3 firmware and tools; these remain compatible with Rev D. Board coordinates below are viewed from the component side, origin at the top-left PCB corner. Bottom silkscreen is read after turning the board over. Do not infer pin numbering from left/right after flipping.

Connector	Pin 1	Pin 2	Pin 3
J1 power	Positive input, 6-22 V	GND	-
J2, J3 CAN	CAN H	CAN L	-
J4 input 1	Selected sensor power	GND	Signal
J5 input 2	Selected sensor power	GND	Signal
J6 input 3	Selected sensor power	GND	Signal

Each terminal pole has two solder legs; they are one electrical contact. J2/J3 are connected in parallel for bus continuation, with no onboard CAN ground terminal. Common ground comes through robot power. Use twisted H/L wiring; keep branches short. Strip 8 mm using the conductor range specified by Weidmüller for the exact terminal. Press the white release button for removal and for conductor types requiring it. Provide cable strain relief independently of the terminal.

Feed J1 from appropriately fused robot distribution. F1 is secondary board protection. Do not backfeed J4-J6 power pins, parallel the two sensor rails, or apply external power to the programming rail. The board is not galvanically isolated.

Voltage jumpers and bias solder settings

The factory ships the board with JP101/JP201/JP301 **unpopulated**. Hand solder the specified 2 mm headers, then leave their caps off for initial bring-up. With power disconnected, fit ONE closed-top cap across pins 1-2 for 5 V or 2-3 for 12 V. No cap disconnects port power.

The three bias selectors and JP2 remain solder bridges with no stencil paste. Use an iron to configure bias/termination; remove the old bridge before changing it.

Channel	Power selector	Bias selector	Center X, mm
1 / J4	JP101	JP103	Power 7; bias 13
2 / J5	JP201	JP203	Power 23; bias 29
3 / J6	JP301	JP303	Power 39; bias 45

The three headers and three bias selectors are at Y=51.5 mm, just above the sensor connectors. **Pads 1, 2, 3 run left to right when viewed from the component side.** The square pad identifies pin 1. Use the detailed assembly drawing/interactive BOM to identify a selector before configuring it.

Selector	Connect 1-2	Connect 2-3	Open
Power: removable cap	Regulated 5 V	Regulated 12 V	Port power disconnected
Bias: solder bridge	10 kΩ pullup to selected port supply, through diode: NPN/open collector	10 kΩ pulldown: PNP	Push-pull or external pullup

One cap per power header, one bridge per bias selector. Connecting all three power pins shorts 12 V to 5 V.

Use Harwin M22-2010305 headers and Sullins SPN02SVEN-RC closed-top caps (2 mm pitch; not 2.54 mm). Caps are friction-retained, not mechanically latched. After voltage/functional verification, support retention with a small removable bead of electronics-grade neutral-cure silicone between the cap exterior and header body, clear of the contacts. This is a proposed retention method; qualify it for your robot vibration/impact environment and enclosure. Allow full cure before use and inspect seating during service. Do not claim these are vibration-qualified merely because there is no slider. Bridge JP2 only at a CAN bus end that needs a 120 Ω termination. Leave it open at intermediate nodes; normally there are two terminations on the entire bus.

Examples:

• Industrial NPN: select the sensor's required supply; bias 1-2. Output current from the pullup is below about 1.3 mA at 12 V. Select inversion in firmware if asserted means pulled low.

• Industrial PNP: select supply; bias 2-3. Verify the sensor's OFF leakage cannot raise the input above the LOW target through 10 kΩ (1.2 V / 10 kΩ ≈120 µA before margin).

• 3.3/5 V push-pull: bias open; select power only if the board powers that sensor. Always connect signal ground.

• 3.3 V open drain that is not 5 V tolerant: bias open, external pullup to the sensor's own 3.3 V rail. There is no onboard 3.3 V sensor supply selector.

Design targets: 0-30 V positive input, LOW ≤1.2 V / HIGH ≥2.4 V, common-ground offset budget ±0.1 V. Do not use as an analog input, quadrature encoder, IO-Link port or continuous negative-voltage input. Sampling is 1 ms with 1-100 ms debounce; short pulses can be missed. Sensor power budgets are shared, 300 mA per rail. Rail-health indications are available; they are not calibrated voltage/current measurements. A fault on either rail makes the health-checked driver input unavailable. See protocol.

Representative industrial sensors

Regulated 12 V is a suitable nominal supply for the following families, subject to exact model and cable checks. Connect the selected output only; insulate unused complementary output wires separately. Brown/blue/black is common supply/ground/output coloring, but follow the specific datasheet rather than assuming colors.

Family	Published limits	Rev D-2 setup / limitation
Omron E3Z three-wire NPN/PNP	12-24 V ±10%; typically 30 mA maximum, through-beam pair 35 mA; residual ≤1 V below 10 mA	Power 2-3 for 12 V. NPN bias 1-2; PNP bias 2-3. Residual voltage fits the LOW target at this board's small load
Banner QS18 standard o pposed/diffuse/retroreflective variants	10-30 V; <25 mA excluding output load; standard outputs <1 V at 10 mA	12 V supply is in range. Select NPN or PNP bias to match model; use one output. Account for its 100 ms power-up delay
Banner QS18 DVS/DL/FF variants	NPN residual specified up to 1.5 V at 10 mA	This exceeds the board's design LOW target. Prefer a compatible PNP variant or qualify actual NPN residual at the ~1.2 mA pullup load; do not assume all QS18 NPN models work

Sources: Omron E3Z specifications, Banner QS18 manual. PNP OFF leakage must also meet the input pulldown budget. Observe manufacturer cable/fusing requirements. Two-wire current-loop, AC, analog, IO-Link and safety/OSSD sensors are not supported as interchangeable three-wire inputs.

12 V is specified at the board, not guaranteed at the remote cable end under every transient. The analytical DC divider/reference budget is approximately 11.80-12.22 V before dynamic/distribution allowances; check ≥10.8 V at an E3Z during startup and worst battery/load conditions. Three 30 mA sensors are comfortably within the 300 mA shared design target, but include emitters, indicator loads, cable losses and inrush. The board's 100 ms rail qualification is not proof that every sensor has completed its own startup.

Programming contacts

J8 uses the **TC2050-IDC-NL mechanical pattern and a custom electrical mapping**, centered at (24, 5.75) mm. Use a mapped adapter, not an assumed ARM/STDC ribbon pinout. Ten 0.7874 mm contact pads are on 1.27 mm pitch, with three mechanical guide holes. In the component-side view, the upper row runs **10, 9, 8, 7, 6** left to right at Y=5.115 mm; the lower row runs **1, 2, 3, 4, 5** at Y=6.385 mm. Pin 1 is at X=21.46 mm, Y=6.385 mm. Use this U-shaped numbering, not an odd/even ribbon numbering assumption.

J8 pin	Signal	Connection
1	NRST	Probe reset, optional
2	MCU UART RX	3.3 V UART adapter TX
3	MCU UART TX	UART adapter RX
4	SWCLK	SWD clock
5	SWDIO	SWD data
6	+3V3	Probe VTref sense **only**, board output
7	GND	Probe ground
8	GND	UART ground
9	BOOT0	Normally leave open; onboard pulldown
10	GND	Ground

Power the board through J1 during programming. Leave adapter/probe power disconnected. BOOT0 is normally low. SWD programming and custom pogo-adapter fit require bench verification. The intended clip-free probe is held perpendicular by a fixture; the no-legs pattern does not retain itself.

4 Hand assembly

Source: docs/ASSEMBLY.md

Factory: fit only the 98 SMT references in Factory_BOM.csv using CPL_JLCPCB.csv. User: buy the nine through-hole parts and three closed-top caps.

Use the exact BOM, assembly drawing, interactive BOM and placement files. Reference labels are in these drawings; dense placement does not leave room to print every reference legibly on the actual top silkscreen.

1. Order the 52 ×64 mm, four-layer, 1.6 mm board with 1 oz finished copper target. Use the exported Gerbers and separate plated/nonplated drill files. Have the fabricator review its actual stackup, annular rings, mask registration and minimum features. Gold finish is useful at the programming contacts; keep solder and paste off them.

2. Check bare-board continuity and orientation. Apply paste to SMT pads only. A 0.10 mm stencil is a starting point; the assembler must review release and the exact QFN apertures. Through-hole header pads, solder selectors, TC2050 contacts and the test point deliberately have no paste.

3. Place 0402/0805/1206/1210 passives, diodes and ICs. Verify C1 polarity, TVS/diode cathodes, pin 1 on each IC and the exact latch-off TPS2553DBVR-1 suffix. U8 TPS552872-Q1 has five long electrical power lands and fine-pitch perimeter pads; it is not a generic single-exposed-pad package. Use its supplied paste pattern, a controlled reflow profile and suitable joint inspection. There are no components on the bottom.

4. Inspect joints and measure for supply shorts. Hand solder the six horizontal Weidmüller terminals after SMT reflow as the user assembly step. Then fit JP101/JP201/JP301: 2 mm headers on the top side, straight and fully seated, with pin 1 at the square pad. Use a removable cap temporarily as an alignment aid with the board unpowered; remove it afterward. Fill both plated holes per terminal pole. Support the board without loading fresh SMT joints. The terminal pins extend below the board.

5. Leave all selectors open for initial bring-up. Supply J1 from a current-limited bench supply, starting at 12 V and a modest current limit. Confirm input polarity, protected input, approximately 5.07 V internal, 3.3 V logic and 12.01 V regulated rails. Sensor headers remain unpowered while power selectors are open.

6. Program the combined factory image through the custom TC2050 mapping, set/verify boot protection, power cycle and configure a unique CAN ID using CAN service. Follow the fixture procedure. Then power off, choose each port's supply/bias and termination settings, inspect the one-cap-per-header settings and bias bridges, and check that 5 V and 12 V are not shorted.

7. Perform the qualification checks before connecting a robot harness. Begin with a known load and known logic source, then the actual sensor/cable.

Use M3 hardware at (3.5,4) and (48.5,42.5) mm. Holes are 3.2 mm NPTH. The copper keepout has at least 3.30 mm radius; use hardware whose contact area stays within a 6.4 mm diameter, with insulating washers/spacers as appropriate. Account for component bodies, connector overhang and protruding pins. Do not clamp the PCB between conductive surfaces. The mechanical drawing gives nominal geometry, not a fitted enclosure.

The compact board still benefits from magnification and a fine iron. Rev D-2 restores removable voltage shunts and retains large passives where possible. See the retention and vibration qualification notes in WIRING.md. The buck-boost QFN and four 0402 capacitors make stencil/reflow work more demanding than Rev C; it is not a through-hole beginner kit.

5 Firmware

Source: firmware/README.md

Bare-metal STM32F042K6T6 firmware for three digital inputs. Build allocation is recorded by the current linker map/build report; stack depth has not been measured. The final two 1 KiB flash pages hold redundant 32-byte versioned configuration with CRC and commit-last writes.

D-2 retains **hardware identity 4** and normal **protocol v3**, with a new service protocol and bootloader. Rev D's 0.4.0 files remain preserved. Use the matching SensorCan12 drivers; voltage selection remains hardware-only.

CAN service is implemented, physically unqualified: full-UID discovery, identify LED, persistent ID/inversion/debounce and verified uploads. See CAN setup/update, robot bridge and qualification plan.

From `rev_d2/`:

```
source tools/env.sh
make -C firmware
bash scripts/test_firmware.sh
```

Outputs: `build/factory.hex` for initial SWD installation; `build/application.sc12` for CAN updates; separate bootloader/application ELF, BIN and MAP files. `sensor_can.hex`/ELF are application-only at 0x08002000. Flash: 8 KiB bootloader, 21 KiB app, 1 KiB manifest, 2 KiB configuration. Both linkers reserve the first 256 SRAM bytes and a 1 KiB minimum stack. No ADC, sensor rail GPIO control or per-input LED logic remains. The 8 MHz external crystal and PLL generate 48 MHz; bxCAN uses 16 time quanta with an 87.5% sample point at 1 Mbit/s. Auto bus-off recovery is enabled; automatic transmit retry is disabled.

Power through J1 from a bench supply. Use the custom TC2050 wiring: VTref sense→6, SWDIO→5, SWCLK→4, ground→7, optional reset→1. Leave probe power disconnected.

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -c 'adapter speed 1000' \
  -f tools/program_factory.tcl
```

This command has not been exercised on physical D or D-2 hardware. Verify MCU identity, supply and flash readback. Normal application programming should preserve the two configuration pages; full-chip erase removes them. Version-1/2 configuration is rejected.

UART: adapter TX→J8.2, RX→J8.3, ground→J8.8; 3.3 V, 115200 baud, 8N1; adapter power disconnected.

```
python host/sensor_can.py info --port /dev/ttyUSB0
python host/sensor_can.py configure --port /dev/ttyUSB0 \
  --device 17 --invert 1 --debounce 2
```

Device IDs 0-63 must be unique in this Team Use address space. Invert is a 3-bit mask; debounce is 1-100 ms. UART `SET id invert debounce_ms` and UID-targeted CAN service save persist and reset. Legacy CAN runtime configuration remains volatile. Unconfigured boards accept discovery/service but suppress normal telemetry. Power-cycle after initial SWD programming and verify boot protection before CAN updates. **Hardware sensor power is enabled independently of firmware**, including during reset or programming. On D-2, leave the caps off JP101/JP201/JP301 to disconnect port power during initial bring-up; configure them only with input power disconnected. Bias and CAN termination remain solder settings. See assembly.

The independent watchdog is serviced only after the full 1 ms foreground cycle. A >20 ms foreground delay, clock failure, hard fault or PVD low-voltage detection puts CAN into standby; the watchdog provides reset recovery. The single LED pulses approximately 100 ms per second when configured and healthy, or 100 ms per 200 ms for configuration/clock/CAN errors. Rail faults also trigger the fast fault indication. PA3 reads 12 V PG and PA4 reads 5 V FAULT_N, both with internal pullups. A fault clears qualification immediately; recovery needs 100 ms continuously good indications. Counts freeze and debounce is re-baselined during this interval.

Broadcast Disable ID 0 stops telemetry and aborts pending transmissions. Local sampling and hardware sensor power continue. An explicit `RESUME` UART command or CAN resume command re-enables telemetry; reads never do this automatically. See protocol v3. Host tests cover firmware logic and mocked peripheral boundaries, not actual oscillator startup, electrical timing or STM32 peripheral behavior.

6 CAN protocol

Source: docs/CAN_PROTOCOL.md

1 Mbit/s classic CAN, extended 29-bit identifiers, 8-byte telemetry. This is an original Team Use protocol, not CTRE Phoenix. Device type=10 (Miscellaneous), manufacturer=8 (Team Use), unique device ID=0-63. Address = `(10 << 24) | (8 << 16) | (API << 6) | device`.

The APIs differ from A/B/B-2; Rev C shares these API numbers but uses protocol v2, which Rev D-2 drivers reject. The version and capability checks prevent drivers from interpreting the wrong revision's payload. Provision each board by full-UID CAN service or 3.3 V UART. An unconfigured board suppresses telemetry but answers service requests; its sensor power hardware is still enabled.

All multibyte fields are little endian. Masks use bits 0-2 for inputs 1-3. Unsupported versions, reserved bits, wrong lengths or stale frames must be reported as unavailable.

API	Period	Eight-byte payload
0x120 status	10 ms	version=3, sequence, debounced input mask, raw mask, channels=3, capabilities=0x3F, flags, reset flags
0x121 counts 1/2	100 ms	two uint32 rising-edge totals
0x122 count 3	100 ms	uint32 total, uint32 reserved=0
0x130 diagnostics	100 ms	uint32 transmit drops, TEC byte, REC byte, LEC byte, CAN flags byte
0x131 identity	1000 ms	3, firmware major=0, minor=4, patch=0, hardware=4, device ID, debounce ms, inversion mask

Status flags: configured=1, clock healthy=2, CAN initialized=4, bus-off=8, qualified 12 V power-good=16, qualified 5 V no-fault=32. Reset flags: power-on=1, pin reset=2, watchdog=4, software=8. Capabilities bits 0-4 indicate edge counts, runtime configuration, UART persistence, standalone hardware power and broadcast disable. Bit 5 indicates rail-health reporting capability. Capabilities are static; flags bits 4/5 carry live qualified indications. The raw mask is the uninverted sampled comparator state. The debounced mask applies configured logical inversion and debounce. Sequence is an unsigned byte and wraps. Counters wrap at 2^32 and clear on reset; runtime configuration preserves totals.

Diagnostic LEC is the STM32 three-bit last-error code (0-7); CAN flags are bus-off=1 and error-passive=2. There is no calibrated sensor voltage/current telemetry. The 12 V PG signal has the converter's typical threshold and tolerances; 5 V no-fault is an overcurrent-switch indication, not proof of voltage. Raw masks remain diagnostic data during faults.

A bad rail indication clears its flag at the next 1 ms sample; restoration needs 100 consecutive good samples. During faults and recovery, firmware re-baselines input debounce and freezes edge totals, preventing supply collapse/recovery from adding counts. All ports require both rail indications because solder-selector positions are not readable. Java/C++ input helpers also require fresh telemetry, configuration, clock health and CAN health. They cannot detect every failed/disconnected sensor or open cable.

Command API	Payload	Effect
0x220 runtime input config	`53 43 03 invert debounce_lo debounce_hi 00 00`	invert 0-7; debounce 1-100 ms; volatile, does not change ID
0x221 resume telemetry	`53 43 03 00 00 00 00 00`	Explicitly resume after broadcast disable

Extended arbitration ID **0** is the FRC broadcast disable. Firmware stops telemetry and aborts pending transmit mailboxes immediately on processing it. Sampling and standalone sensor power continue; there are no actuator outputs. Resume requires the explicit command above, UART `RESUME`, or reset. Drivers never resume from a read loop. Integrate explicit resume with a reviewed robot lifecycle policy; this implementation still requires robot-bus testing and does not establish competition approval. See WPILib CAN addressing.

Read timeouts: status 100 ms, counters/diagnostics 350 ms, identity 1500 ms. Counters 1/2 and 3 are separate snapshots. Sampling is 1 kHz; debounce requires a continuously stable logical state for the selected interval. This is not a high-speed pulse counter or encoder.

UART: 115200 baud, 8N1, 3.3 V logic. Commands `INFO`, `HELP`, `SET id invert debounce_ms`, `RESUME`. `SET` saves a version-3 redundant CRC-protected record and resets. The old four-argument command and v1/v2 flash records are rejected; provision Rev D-2 explicitly. No power-control command exists.

Service extension in firmware 0.5.0

Normal protocol-v3 payloads remain unchanged. APIs 0x300/0x301 at address 0 are reserved for CAN service. Opening maintenance pauses telemetry/counting on participating SC12 nodes; treat samples as unavailable immediately in robot code. Status period can be 10/20/50 ms until reset.

7 Robot integration

Source: examples/README.md

Java 17 and C++ examples target WPILib 2026.2.1. D and D-2 share protocol **3**, hardware **4** and the same sensor driver interface. D-2 adds service firmware **0.5.0**; D retains **0.4.0**. D-2's passive substitutions and removable voltage jumpers do not change the CAN messages. Rev C Lite frames and the earlier four-input protocols are rejected. Copy the relevant `SensorCan12` and `SensorCan12Codec` files into a robot project, then set the same unique device ID provisioned through CAN service or UART.

Choose each sensor's 5 V / 12 V supply using the physical jumper before operation; the drivers do not select port voltage. See the D-2 setup guide and wiring.

CAN discovery, persistent settings and firmware updates are provided by the new service tools. The robot bridge has separately compiled 2026 and 2027 alpha-7 adapters. Sensor driver classes below remain pinned to 2026.2.1; the 2027 bridge does not port the normal drivers. Physical end-to-end CAN testing is pending.

```
private final SensorCan12 sensorBoard = new SensorCan12(17);
// In periodic(): absent means unavailable or a reported fault.
var detected = sensorBoard.input(1);
if (detected.isPresent()) {
  boolean sensorActive = detected.get();
  // Use a reviewed subsystem policy for this observation.
}
```

C++: `sensorcan12::SensorCan12 board{17};` then `board.ReadInput(1)` returns `std::optional<bool>`. Construct the driver once; an empty result must not silently become a valid “no object” observation. The helper requires fresh status, configured identity state, healthy clock/CAN and both qualified power indications. Raw `status()` / `ReadStatus()` remains available to diagnose faults. Counts are frozen during power faults and wrap at 2^32; compare/reset policy in the robot application.

Reads do not transmit or resume telemetry. `configureRuntime` / `ConfigureRuntime` sends volatile input inversion/debounce; it does not set persistent ID. `resumeTelemetry` / `ResumeTelemetry` is an explicit action after investigating broadcast disable, not a periodic automatic retry. Status timeout is 100 ms; count/diagnostic timeout 350 ms; identity timeout 1500 ms.

The Python host tool supports UART `info`, `configure`, `resume` and passive CAN `monitor`. UART uses 3.3 V levels at 115200 8N1; disconnect adapter power. CAN monitor requires a suitable external CAN interface; a UART adapter is not a CAN interface. Run the following from `rev_d2/`:

```
source tools/env.sh
python host/sensor_can.py configure --port /dev/ttyUSB0 --device 17 --invert 1 --debounce 2
python host/sensor_can.py monitor --interface socketcan --channel can0 --device 17
bash scripts/test_drivers.sh
```

Tests cover codec boundaries, all reserved status bits, stale/malformed responses, fault rejection, unsigned counts and explicit-only writes. The C++ example compiles against the actual pinned WPILib headers; Java compiles against pinned WPILib jars. Hardware bus behavior and competition acceptance remain open qualification steps. See protocol, wiring, and validation.

8 Purchasing and factory quote

Source: docs/FACTORY_ORDER.md

Use RevD2_SC12_Factory_Upload.zip for a **prototype lot**, after reviewing the validation plan. No board has been physically qualified, no factory quotation is included and no order has been placed. The exact-MPN BOM is intended for a Chinese or other assembler that accepts customer-specified sourcing; it is not a pre-approved JLCPCB/LCSC turnkey basket.

For a JLCPCB upload, start with the fabrication-only Gerber ZIP, then supply BOM_JLCPCB.csv and CPL_JLCPCB.csv. Keep Factory_BOM.csv alongside for exact-MPN matching. The larger factory packet is a complete reference handoff.

Fabrication

Setting	Requirement
Board	52 × 64 mm, four copper layers, FR-4, 1.6 mm nominal
Layer order	F.Cu signal/power, In1.Cu GND, In2.Cu GND, B.Cu signal/power
Copper	35 µm nominal each layer; verify finished copper with supplier
Dielectric	Symmetric four-layer stack; target outer-to-inner dielectric ≤0.25 mm; supplier must confirm final stack
Finish	ENIG specified for exposed programming contacts and fine-pitch QFN
Minimum trace	0.18 mm
Clearance	0.20 mm general; U8 package pad-to-pad 0.15 mm local minimum
Vias	Through vias, minimum 0.30 mm drill / 0.60 mm pad; no blind/buried vias
Drills	Separate PTH and NPTH Excellon; no plated mounting or alignment holes
Solder mask	Both sides; preserve fine-pitch openings; review U8 mask webs with CAM
Stencil	Top only, nominal 0.10 mm laser-cut; use supplied paste apertures
Panels	Factory rails/fiducials may be added outside the delivered outline; preserve connector overhang

Native nominal stack: 0.035 mm copper on all four layers, 0.20 / 1.04 / 0.20 mm dielectric and 0.01 mm solder mask per side, totaling 1.60 mm. These are planning thicknesses; final fabrication tolerances and material selection belong to the approved supplier stack.

This stack is an order requirement, not a confirmed laminate/impedance quotation. CAN is short board-level routing; no fabricated impedance coupon has been qualified. Get the supplier's CAM/stackup review before fabrication.

Assembly

Factory scope: 98 SMT components on the top side only. Use `bom/Factory_BOM.csv` (exact MPNs) and `output/assembly/CPL_JLCPCB.csv`. The simplified `bom/BOM_JLCPCB.csv` is a matching aid; blank LCSC numbers require exact-MPN matching against Factory_BOM, never an automatic substitution.

User scope: six terminal blocks plus three voltage headers (nine through-hole parts), then three loose caps. The factory must leave J1-J6 and JP101/JP201/JP301 empty. `positions_all_top.csv`, the complete BOM and drawings describe the finished board, not additional factory placements.

CPL coordinates are mm from the PCB lower-left, positive X right and Y up, with native KiCad counterclockwise rotation. Match the uploaded preview to the assembly drawing and check IC/diode pin 1; machine-library rotation conventions can differ. R64 is physically a normal two-terminal 1206 resistor; its four numbered PCB

pads include two copper-only Kelvin taps. NT1 is a copper ground tie and is not a purchased component. Do not interpret pad count as component termination count.

Power headers are unpopulated; all bias/termination solder bridges and all J8 contacts are **unbridged/unpasted as supplied**. The factory must not add solder across selectors unless a separate reviewed port-configuration work instruction is supplied. The user will hand solder all through-hole parts after delivery. The family name “LSF-SMT” does not make these exact connectors ordinary two-pad SMT parts.

Do not replace U8 with a generic 21-pin QFN, substitute a lower-saturation inductor, change the precision divider grade, or replace the 1210 capacitor bank with a nominally equal capacitance without approval of the actual electrical/footprint data. DigiKey substitute labels alone are insufficient. Maintain 0.1 ohm shunt value, tolerance and terminal orientation.

Include inspection of fine-pitch joints and the five long QFN power pads with suitable inspection/X-ray for the prototype lot. Perform unpowered shorts/selector tests, current-limited staged bring-up, firmware programming/readback and functional rail/input/CAN tests before accepting units. Track lot and component substitutions.

Sourcing, cost and delivery

Cost reduction and batch estimates separates parts procurement from assembly. LCSC inventory is not proof of JLCPCB factory allocation. No SMT part allocation, complete cart or final quotation has been confirmed. Request exact-MPN matching and incoming date confirmation before paying; the STM32 and TPS552872-Q1 may require outside procurement. Any unapproved item must remain unresolved rather than being silently substituted.

Request Economic PCBA first if the exact parts and four-layer board qualify; compare Standard PCBA with the same SMT-only scope. Avoid paying for bottom-side assembly or through-hole work. Standard assembly still charges feeder loading for all 46 SMT types. No programming or functional-test service is included by this package unless separately quoted; the user performs bring-up and programming.

A delivery within one month is plausible with allocated parts and tracked air express. Budget 2-4 weeks as a planning allowance, not a carrier promise; procurement, review holds and customs can exceed it. Country/postcode and order quantity are needed for an actual delivered quote. Have the manual parts arrive in parallel. Do not select a long supplier lead time to obtain a lower component price when the one-month target matters.

Programming deliverables

The packet includes combined `firmware/build/factory.hex` and `tools/program_factory.tcl` for initial SWD programming/protection. Power cycle and verify protection, then use CAN for setup and later `application.sc12` updates. User programming is the default quoted scope; see fixture procedure. Do not substitute an application-only ELF/HEX.

9 Validation and qualification

Source: docs/VALIDATION.md

No Rev D-2 prototype has been assembled, powered or tested. The files support engineering review and prototype fabrication. The production/competition qualification gates below remain open, even when every automated check passes.

Automated evidence

The release contains native KiCad electrical/design-rule reports, explicit pin-model/schematic/PCB parity, placement/selector/programming-contact checks, manufacturing-file parsing and copper checks, host firmware tests, and Java/C++/Python protocol tests. These check design consistency and modeled behavior. Mocked peripheral tests do not prove STM32 timing or CAN electrical operation.

Electrical checks sweep input-threshold tolerances, divider accuracy, resistor stress, fuse/load budgets and manufacturer-curve capacitor derating. The separate converter model screens boost CCM loop corners only. Neither substitutes for hardware measurement. Reports state their assumptions rather than treating typical values as guaranteed bounds.

Prototype gates before production release

Gate	Measurement / acceptance intent
Assembly	Confirm exact horizontal terminals, pin-to-hole fit, polarity, U8 long-pad joints, R64 Kelvin layout, selector isolation and no solder shorts
Staged power	No sensors first; current-limited 12 V source, verify 5 V/3.3 V before full operation, then regulated 12 V and low-load input current
Regulation	Sweep 6-22 V with 0-300 mA on each rail separately and together; verify 12 V remains within the selected sensors' supply limits at their cable ends
External bias	Scope V5_INTERNAL, BB_EN, V12_RAW and 3V3 during slow/fast input ramps, repeated starts and brownouts. Confirm converter bias never causes unsafe switching or back-powering; review operation below 6 V
Transients	6↔14↔22 V line steps; 0↔300 mA load steps; representative sensor startup/inrush and cable lengths; check ringing/undershoot through the buck/boost crossing
Stability	Measure loop response or justified equivalent transient qualification in buck, boost, transition and light-load/PFM modes; target >45° phase and >10 dB gain margin where measured
Faults	Short each sensor supply and each port; verify current-limit behavior, 12 V automatic recovery versus 5 V latch, firmware flags, no spurious counts and safe temperatures. Do not assume every 12 V short enters hiccup
Electrical protection	Reverse input, realistic protected-branch fault energy, fuse clearing/I^2t, inductive transients and TVS stress using suitable test equipment. Not automotive load-dump qualified
Temperature	Both rails fully loaded at 6/12/22 V, intended ambient and later enclosure; measure U8/U1/L1/L2/Q1/fuse/terminal and MCU temperatures, hot-start/restart and surface safety
Inputs	0-30 V PNP/NPN/push-pull thresholds, off-state leakage/residual voltage, ground offset, sensor disconnection, wiring transients and all intended sensor models
CAN/firmware	Real roboRIO bus at 1 Mbit/s with correct end termination; no normal telemetry before provisioning, unique IDs, bus-off, clock faults, watchdog, broadcast disable, explicit resume, stale data and power-fault handling
EMC/mechanical	Motor-noise environment, ESD at terminals, cable retention, vibration, mounting insulation and programming-adapter fit; repeat after enclosure design
Production process	Approved stackup/CAM, stencil/profile, supplier alternates, test fixture, serial/lot records, programming/provisioning and final functional limits

FRC integration

Use a single appropriate protected power-distribution branch and wiring sized for its protection. The 0451 3 A fuse has a published 300 A interrupt rating at 32 V DC; confirm prospective branch fault current and clearing energy for the actual harness. The onboard fuse is secondary protection, not permission to bypass the robot distribution rules. Sensor grounds are common with the board; isolate conductive sensor cases where required by frame-ground rules. Do not use dedicated radio/roboRIO power outputs as a convenient supply.

The custom Team Use CAN address uses manufacturer 8 and miscellaneous device type 10. The implementation processes broadcast disable ID 0; it has no actuator outputs. It is not a CTRE Phoenix device. Robot inspection and the current season's rules still apply. Commercial CAN product acceptance, assigned identifiers and required ecosystem support are separate outstanding steps; Java/C++ examples alone do not establish approval.

Primary references: 2026 FIRST manual, especially custom-circuit voltage, distribution/wire and frame-ground rules; WPILib CAN addressing and acceptance requirements. Recheck applicable rules for the actual event/season.

D-2-specific prototype checks

Verify the exact 2 mm header and cap fit, both 5 V / 12 V positions and unpowered open state on every channel. Verify no connection between the two rails in either position. Check solder fill on all 39 user-fitted through-hole joints. Fit/retention, vibration and impact testing must include the actual caps, chosen removable retention compound, enclosure and robot harness; these headers have no positive latch. Compare the substituted passive values/ratings against the supplied BOM. Factory BOM and CPL must contain exactly 98 SMT references, with all nine through-hole components omitted. Firmware identity intentionally remains hardware 4 / protocol 3.

CAN servicing qualification

The service engine and host uploader pass simulated flash cuts/retries; ARM register checks and both bridge adapters compile. Before robot use, measure actual boot/vector/interrupt behavior, option-byte protection, stack use, upload time, CAN load, bus-off recovery and duplicate-ID discovery. Interrupt power during invalidation, every erase/program phase and final commit on physical units. Verify persistent configuration, bootloader recovery, controller disconnect/enable rejection and healthy telemetry after exit. Protocol and evidence; per-unit fixture records. No software simulation closes these physical gates.

10 Connector evidence

Source: docs/CONNECTORS.md

References	Exact MPN	Series description	DigiKey
J1-J3	**1824420000**	LSF-SMT 3.50/02/90 3.5SN BK TU	281-2020-ND
J4-J6	**1824430000**	LSF-SMT 3.50/03/90 3.5SN BK TU	281-2021-ND

Manufacturer pages: 2-pole, 3-pole. `/90` gives horizontal wire entry parallel to the PCB. `/135` is the angled version and is not used. Phoenix PTSM is not fitted.

These are through-hole/THR parts despite “SMT” in the family name. Pitch is 3.5 mm, with two solder legs per pole and 7 mm row spacing. PCB drills are 1.1 mm. Housing widths are 7.7 mm / 11.2 mm. Manufacturer CAD is 14.3 mm deep; the footprint conservatively reserves 14.75 mm. Height is 8.5 mm above the mounting surface; nominal pin tips reach 3.5 mm below that surface (about 1.9 mm beyond a 1.6 mm PCB).

Current renders use the official STEP solids with a rigid translation (-2.13,-12.625,-3.5) mm into the footprint origin. They are not approximate connector boxes. Display colors are assigned by the project. Original files, transformed files, URLs, hashes, pin/drill checks and nominal bounds are in vendor_connectors.json. Script `import_connector_models.py` checks all six placed pad patterns and the two vendor models. Do not regenerate models merely to change routing: STEP exporters can alter file headers without changing geometry.

Top terminals face outward at the power/CAN edge; three sensor terminals face outward at the opposite edge. Conservative complete footprint envelope is 52 ×72.1 mm; add room for cable bend, strain relief and button access. Renders do not validate conductor retention, tool clearance in an enclosure, molding tolerances or actual solder fit.

Rev D-2 voltage headers

JP101/JP201/JP301 use Harwin M22-2010305: 2 mm pitch, 0.50 mm square pins, 6 × 2 mm body, nominal 5 mm height above the PCB and 3.4 mm tails. The project’s 2 mm header footprint uses 0.8 mm drills and a 0.25 mm nominal body courtyard margin. Pads 1-3 run left-to-right at Y=51.5 mm; pin-1 positions are X=5, 21 and 37 mm.

The Sullins SPN02SVEN-RC drawing specifies a closed top, 2 mm pitch, 0.50 mm square mating pins and nominal 4 × 2 × 4.92 mm body. Contact height is 2.45 mm from the entry, within the header’s 3.5 mm mating pin length. Nominal mating geometry agrees; physical fit and retention remain untested. Caps are delivered loose and not included in the assembled-board STEP model. Use the finished cap envelope for enclosure access/clearance and allow room to remove them.

The generic local header STEP represents the nominal pin/body geometry, not manufacturer-qualified tolerances. Vendor connector hashes continue to cover the unchanged six terminal models. See wiring for one-cap-per-header configuration and vibration retention qualification.

11 Tools and reproduction

Source: TOOLS.md

The saved native KiCad schematic/PCB are authoritative. Generation/placement/router scripts record design construction; they are not a one-command reroute of the reviewed board. Editing the circuit requires matching schematic, BOM, PCB and new checks. Earlier revisions remain separate.

Tested toolchain: KiCad 10.0.6, its matching Python/pcbnew interpreter, Python 3.12, Arm GNU GCC 13.2.1, GNU make, host GCC/G++, Java 17, Poppler, InteractiveHtmlBom 2.11.2 and the packages pinned in `tools-requirements.lock`. Tool binaries/caches are external to the repository. Install those native tools for the host operating system. The `kicad-python` wrapper must use the interpreter matching the KiCad binary bindings, not an arbitrary venv Python.

```
# In rev_d2/, after installing native tools:
uv venv --python python3.12 .venv
uv pip sync --python .venv/bin/python tools-requirements.lock
source tools/env.sh
python tools/fetch_driver_dependencies.py
bash scripts/validate_release.sh
```

`tools/env.sh` uses a revision-local venv, or the repository venv when working in the full checkout. It prepends `$CANDI_TOOLS_ROOT/bin` (default `$HOME/.local/share/candi-tools/bin`). Wrappers expected there are `kicad-cli`, `kicad-python`, `host-gcc`, `host-g++`, `ibom`, and Arm compiler commands. Java is under the configured tools root or selected with `CANDI_JDK`. Fonts for the PDFs are DejaVu Sans/Mono.

The validation script checks the saved design and refreshes manufacturing evidence, firmware and host tests; it does not move or reroute components. A visual/documentation refresh is separate:

```
bash scripts/render_board.sh
python scripts/build_documents.py
python scripts/build_handbook.py
python scripts/review_documents.py
python scripts/build_release.py
```

The factory packet is built by `build_documents.py`. Rebuild it after any BOM/placement/fabrication change. `build_release.py` selects authoritative files and writes a manifest plus ZIP checksum. Check an extracted archive with `verify_archive.py`; the release manifest deliberately excludes itself and the enclosing ZIP from its file hashes.

Use `python scripts/import_connector_models.py` only when restoring/reviewing vendor models. It checks the original STEP hashes and all six native connector pad patterns, then writes translated local models. It can change STEP headers, requiring a fresh STEP export, renders and release. Keep the originals and provenance JSON together.

`hardware/design/circuit.py` is the explicit component/net model; `generate_schematic.py` rebuilds the schematic. Construction helpers such as `pack_compact.py`, `route_critical.py` and `apply_routes.py` operate on intermediate files and can overwrite placement/routing. They are not part of release validation. The boost CCM analytical loop screen is in `check_converter_loop.py`. It is not a switching simulation or thermal qualification. Do not reuse older revision reports as evidence for this changed supply.

In the full repository, `kicad-python scripts/check_variant.py` compares D-2 component pin nets, passive values/footprints, critical routes, and records the intentional firmware divergence with the preserved neighboring `rev_d/`. The dated result is included in variant_verification.json; a standalone D-2 archive does not contain the older D baseline.

For CAN servicing also run `CANDI_JDK25=/path/to/jdk25 bash scripts/test_bridge2027.sh`. Dependencies are pinned in `tools/bridge2027_dependencies.json`; Java 25 is required by 2027 alpha-7. `test_firmware.sh` includes C upload-engine fault injection and compiled ARM register tests. `compare_assembly_cost.py` generates partial-fee scenarios, not live quotes.

12 Sources and notices

Source: docs/SOURCES_AND_LICENSES.md

KiCad library/model notices are retained under `licenses/`, together with Arm CMSIS, ST device-header and WPILib notices. Firmware third-party directories and InteractiveHtmlBom output retain their notices. Original project work has no selected blanket license. Manufacturer documents/CAD remain their owners' material; retaining a file is not a redistribution-license grant.

Subject	Primary reference
Input fuse interruption / derating	Littelfuse 451/453
Buck and capacitors	TI LMR51420
Regulated 12 V converter / exact RYQ0021B package	TI TPS552872-Q1, SLVSHF9 August 2023
Shielded 3.3 µH inductor	Bourns SRP5030TA
Input/output/bias capacitor curve	Samsung CL32B106KBJNNW
Banner supply/residual voltage exceptions	QS18 product manual
5 V latch-off limiter	TI TPS2553, -1 variant
Comparator/leakage caveats	TI LM339LV
CAN interface / bypass	TI TCAN3413
STM32 pinout, supplies, clock	ST STM32F042
Example industrial output residual levels	Omron E3Z specifications
Terminal geometry and exact MPNs	Weidmüller 1824420000, 1824430000
CAN addressing / broadcast disable	WPILib

Samsung CL31A226KAHNNNE typical characteristic data is retained in `sources/datasheets/samsung_CL31A226_characteristics.json`, inherited from the earlier design's manufacturer retrieval. Typical curves do not establish guaranteed capacitance. DigiKey URLs, exact codes and dated observations are in `bom/catalog_selected.json`. Vendor STEP URLs, hashes and transforms are in `hardware/library/3d/vendor_connectors.json`.

Equivalent parts need engineering review; distributor substitution suggestions do not establish electrical/mechanical equivalence. Availability and season rules can change after these observations.

Rev D-2 retains the retrieved Samsung CL32B106 characteristic JSON and TI converter datasheet under `sources/datasheets/`. The project-created converter footprint/model is documented in `hardware/library/converter_package.json`; its STEP representation is parametric package geometry, not official manufacturer CAD. The horizontal connector STEP models are official vendor geometry.

D-2 sourcing URLs and tiers are in catalog_china.json. Header/cap geometry links the Harwin and Sullins manufacturer drawings; cost assumptions links the assembly fee and lead-time sources. Stock observations are not factory allocations.

13 Cost reduction

Source: docs/COST_REDUCTION.md

Pricing basis: USD supplier observations collected 2026-09-07 through 2026-09-08. These are parts budgets, not a factory quote or a reserved basket. Some unchanged-part observations are carried forward from the D/B-2 catalogs. The selected sources and price breaks are recorded in catalog_china.json.

Implemented changes

1. Sixteen ordinary 0805 resistors change to same-value UNI-ROYAL 1%, 125 mW equivalents; eleven 100 nF 0805 bypass capacitors change to Samsung CL21B104KCFNNNE, X7R, 10%, 100 V (formerly 50 V). Footprints and electrical values remain unchanged. Precision resistor grades, power ratings, bulk banks, converter bypass capacitance/compensation values and protection are retained. Per-reference substitutions.

2. Unchanged active/power MPNs use lower supplier observations when available: these are procurement changes, not unreviewed circuit substitutions. LCSC candidates include the fuse, reverse-polarity MOSFET, 5 V switch, buck, CAN transceiver and selected capacitors.

3. Factory scope is only 98 SMT placements on the top side. The user fits six terminals plus three voltage headers and supplies three closed-top caps. This avoids factory through-hole labor and connector procurement while retaining the existing connector geometry and quality. It does not eliminate the cost of buying connectors.

4. Three removable 2 mm voltage selectors replace D’s permanent solder settings. No slide switches or bottom-side components are introduced. Four copper layers remain: assembly side count and copper layer count are separate choices.

There are still **46 SMT MPN types**. Substituting cheaper resistors lowers component cost but does not by itself reduce feeder count or establish Basic-part status.

Batch parts budget

Boards	Parts purchase cash per board	Parts purchase cash for lot
1	$55.27	$55.27
5	$40.40	$202.01
10	$34.69	$346.87
25	$32.23	$805.83

Includes all SMT parts, six terminal blocks, three headers, three caps, and distributor minimum-order/multiple rounding. At ten boards, SMT purchasing is $26.10 per board and manual parts/caps are $8.59. The calculation minimizes purchase cash across MOQ/multiple rounding and all published price breaks; details show every line and leftover quantity. No attrition allowance is included because the factory’s requirements are unknown.

D’s $54.89 figure was a quantity-one consumed-parts estimate. D-2’s consumed-parts figure at each supplier’s first saleable tier is $44.16, about 19.5% lower, including the new headers/caps. Buying distributor minimums for just ONE D-2 board instead costs $55.27 in this model. **A single board is not a good cost target with this BOM.** Batch pricing is the larger saving; comparing D’s one-board figure to D-2’s ten-board figure would exaggerate the substitution savings.

PCB fabrication, stencil, setup/feeders, placement, inspection, sourcing fees, attrition, programming, functional testing, freight, duties/taxes and your assembly time are excluded. The exact factory unit price may differ from these public distributor prices.

Assembly allowance

JLCPCB's published assembly fees, updated 2026-08-20, separate Economic and Standard service. For this one-sided board, Standard's $25.56 setup, $8.21 stencil and 46 × $1.53 feeder loading total **$104.15 per order** before placement and other charges. Their two-fixture allowance for 1-29 boards is another $16.42. At ten boards these two items alone allocate $12.06 per board.

Placement, leadless-package X-ray, packing and quote-specific adjustments add to that. Hand soldering avoids the published $3.58 order labor charge plus $0.0164 per manual joint (39 finished-board joints per board), if those charges otherwise apply. It is a useful process simplification, not a $10-per-board saving by itself. Confirm the factory's final scope and fee calculation.

Economic service has lower setup costs but charges extended-part feeder fees and depends on exact inventory/eligibility. Do not budget zero extended fees until every factory part is matched. Request both quotes for the same SMT-only scope. With ten boards, roughly $35 in purchased parts plus about $12 in Standard setup/feeder/fixture allocation already approaches $47 before other assembly charges, bare PCB and freight. A delivered sub-$50 board is therefore **not established** by this revision.

Remaining major costs

At ten-board quantities the six specified Weidmüller terminals are about $7.85 per board; TPS552872-Q1 is $5.42 and STM32F042K6T6 is $3.28. Together they account for about $16.55 before the inductor, regulators, protection and passives. Large further savings would need a reviewed connector or power/MCU redesign, or a materially better factory quote; passives alone cannot remove those costs.

The public listings used include Weidmüller 1824420000, 1824430000, TPS552872-Q1, STM32F042K6T6, Samsung bulk capacitor, and selected 100 nF capacitor. Sources can change without notice.

China sourcing and one-month delivery

Start with an exact-MPN SMT quote and the factory package. The JLCPCB MCU listing identifies a library part, but does not confirm allocation for this order. MCU and converter procurement are potential schedule bottlenecks. LCSC stock is distinct from JLCPCB factory inventory. The observed bulk-cap stock was only 220 pieces against seven fitted per board; larger lots plus attrition need fresh confirmation.

JLCPCB's lead-time explanation describes assembly time after bare boards are ready and explicitly calls out review and component-preparation dependencies. Fast SMT alone is not the complete delivery time.

Planning judgment: delivery within a month is plausible with all parts allocated, prompt review and tracked air express. Allow 2-4 weeks for planning, subject to a destination-specific quote; procurement delays, customs or production holds can push it beyond a month. Order quantity and destination/postcode are still unspecified, so no shipping price or arrival date is claimed. Buy the manual components in parallel. No order, payment or supplier message has been sent.

Further optimization delivered with CAN service

The purchasing script checks whether buying to the next price break costs less in total. At five boards, three precision resistor lines are cheaper as ten-packs, saving $0.48 per order ($0.096/board). Ten-board purchasing remains $34.69/board. No components were removed or changed for firmware 0.5.0; the bootloader fits the existing MCU.

Assembly scenarios compare parts with published setup/stencil/feeder/fixture fees. These are partial subtotals, not quotes. For 5/10/25 boards, Standard subtotals are $64.51/$46.75/$37.05 per board before PCB, placement, inspection, attrition, handling, programming/test, freight and tax. Economic is lower on this restricted basis when no more than 36 types incur its extended feeder charge. With all 46 charged, Economic is higher; confirm Basic/promotional status and service eligibility.

The exact 1824420000 LCSC listing reviewed shows zero stock; its reference price is not counted as available savings. A latched connector redesign must include housings/contacts/crimp labor and vibration qualification. Keep the circuit/terminals until a reviewed replacement and complete quote justify a new hardware revision.

14 Optimization progress and qualification

Source: docs/OPTIMIZATION_PLAN.md

Updated 2026-09-08: D-2 now delivers application 0.5.0 and bootloader 1.0.0, CAN commissioning/updating, laptop CLI/GUI and robot bridge adapters. Software tests pass; the board and controller workflow remain physically unqualified. Earlier planning/binaries are preserved in git at `7078fd9`; the neighboring Rev D remains firmware 0.4.0.

What was implemented

Work	Current evidence	Remaining gate
Separate bootloader/application	Fits existing 32 KiB STM32; protected 8 KiB boot region, 21 KiB app, manifest, redundant configuration	Real option-byte programming, handoff/interrupts and stack high-water
Discover/blink/persistent settings	Full 96-bit UID selection, readback, duplicate-ID simulation, explicit save/reset	Several physical blank and duplicate-ID boards; power-cycle verification
CAN update/recovery	Actual C engine tested through uploader; 2,148 simulated erase/program cut points, retries, bounds/CRC checks	Physical power cuts, bus-off and actual CAN timing; no rollback
Laptop and controller workflow	Python CLI and Tk GUI; real TCP fragmentation/revocation test with mock CAN; 2026 and 2027 alpha-7 adapters compile	End-to-end roboRIO/Systemcore and desktop adapter testing
Programming/test process	Combined factory image, protected programming script, per-UID acceptance-record tool and measurement template	Build fixture; measure pilot units and programming/test labor
Cost	Exact existing MPNs retained; purchase-quantity optimizer and assembly-fee sensitivity table generated	Allocated exact-MPN quotes, quantity/destination, lead time and full delivered price

Use CAN service for protocol and operating instructions, fixture for unit records and cost reduction for current numbers. Compilation and simulations do not close the physical gates. No PCB redesign or new bill-of-material component was needed for servicing.

Lessons taken from LaserCAN and GrappleHook

LaserCAN separates its application and CAN bootloader while retaining SWD for initial installation/recovery. Its STM32F103 bootloader reserves 20 KiB and uses different vector handling; SC12 implements its own compact Cortex-M0 C engine and SRAM remap. Sources: pinned bootloader, memory map, SWD instructions and ST RM0091.

GrappleHook demonstrates serial-based identification/settings, acknowledged offset uploads and a TCP-to-CAN robot path. SC12 provides those architectural capabilities with its own protocol and UID discovery. Our bridge remains inside running robot code, requires opt-in plus disabled state, and binds localhost for SSH access. Reference: pinned management/upload code and roboRIO provider. No reference implementation code was copied; both repositories carry GPL-3.0 notices. GrappleHook does not recognize SC12.

SC12 continues to serve three external industrial digital inputs. LaserCAN's integrated ranging sensor has different power/interface needs, so its board size and price are not like-for-like targets. SC12 retains regulated 5 V/12 V at 300 mA shared per rail, 6-22 V input target, top-side-only assembly, removable power caps and no slide switches.

Next qualification loop

1. Program two pilot boards through SWD; verify protected pages and configuration retention. Measure boot/startup, watchdog recovery, vector interrupts, stack headroom and update time. Confirm the firmware truly returns to healthy telemetry after maintenance.

2. Test duplicate IDs, blank devices and mixed boot/application states on a real CAN network. Disconnect controller/host, interrupt power at erase/program/commit, corrupt images and force bus-off. Repeat uploads and verify configuration and bootloader integrity. Verify the controller bridge's permission loss and robot output interlock.

3. Exercise representative Banner/Omron models at cable ends: both rails independently/together to 300 mA, 6/12/22 V input, startup, shorts, brownout and enclosure temperature. Validate sensor residual-voltage limits rather than assuming all NPN sensors work.

4. Qualify terminal, mounting, harness and actual cap/retention compound under vibration/impact. An absent cap or disconnected sensor can resemble a valid inactive input; do not claim disconnect detection. Consider a passive CAN bypass harness for board replacement without opening the robot backbone.

5. Record all observations per UID and lot, correct failures, repeat affected tests and publish the measured limits. Complete the validation matrix before production use.

Cost decisions

Keep the reviewed circuit for the pilot. The software fits the current MCU without external memory or a USB connector. The small implemented purchasing correction saves $0.48 across five boards by buying cheaper ten-packs for three precision resistor MPNs; it does not materially change the ten-board price. Parts remain $34.69/board at ten and $32.23 at 25, before assembly/PCB/shipping.

The useful next step is an exact-MPN quote comparison at 5/10/25 boards. Standard fees charge all 46 SMT types; Economic charges extended types differently. The scenario CSV keeps unknown fees separate, with no assumed free feeders. At 25 boards Standard's parts-plus-setup/feeder/fixture subtotal is about $37.05/board versus $46.75 at ten, still excluding fabrication, placement, inspection, freight and other charges. Buying more boards lowers unit allocation while increasing cash for the order.

Connector redesign remains a candidate, not a substitution: compare complete latched-crimp harness cost including housings, contacts, crimp tools/labor, spare harnesses and strain relief. Verify mating cycles, wire gauge, current, polarity, footprints and vibration. Current terminals cost about $7.85/board at ten; the cheaper exact-MPN LCSC listing reviewed had zero stock and is not a basis for promised savings or delivery.

A converter redesign must compare the complete power circuit and requalify regulation through battery sag, thermal behavior and protection. Keep four copper layers and the critical switching/return layout until an actual quote demonstrates worthwhile savings. Preserve effective bulk/local capacitance and precision compensation values. A larger MCU or CAN-FD controller requires a separately justified feature and new hardware revision.

China delivery inside a month remains conditional on allocated components, review/manufacturing time, destination-specific air shipping and customs. No order has been placed, no supplier contacted and no delivered price/date confirmed by this work.

15 CAN setup and updates

Source: docs/CAN_SERVICE.md

D-2 firmware **0.5.0**, bootloader **1.0.0**, service protocol **1**. Normal sensor messages remain protocol **3**, hardware **4**. Implemented and software-tested; actual STM32/robot-controller, power-cut and vibration qualification is pending. This is our own service protocol, not GrappleHook-compatible firmware.

First installation and recovery

Install factory.hex once through SWD using program_factory.tcl. This combines the protected bootloader, relocated application and image manifest. Existing 0.4.0 boards need this one-time SWD migration; they cannot install it over CAN. The script preserves the two configuration pages and refuses an unexpected device/flash size or an already protected bootloader. Its hardware execution is a pilot qualification gate.

From `rev_d2`, with J8 wired per the custom mapping and a separately powered, current-limited board:

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -f tools/program_factory.tcl
```

Power cycle afterward and verify `boot_protected: true` using the tool. Write protection covers flash pages 0-7 (OpenOCD protection blocks 0-1). Updating requires those blocks locked and application/configuration blocks writable. The script does not change readout protection or mass-erase configuration. Do not program `sensor_can.elf` or `.hex` alone as an initial install: these are relocated application-only files without the boot-valid manifest. Do not upload `factory.hex` over CAN.

Normal CAN updates use application.sc12. If interrupted, retry the complete file: there is no old-image rollback or cross-session resume. Invalid images remain in the bootloader. A normal boot gives a 500 ms discovery opportunity; an explicit boot request stays there, and three successive watchdog resets before two healthy application seconds also request recovery. `run` exits a valid bootloader image. SWD remains the recovery path for bootloader/option-byte damage or a nonfunctional CAN connection.

Laptop and robot use

Use Python 3.10+; the robot tunnel transport needs only the standard library. SocketCAN additionally needs `python-can`; configure the adapter for classical CAN at 1 Mbit/s. Tk is required only for the desktop window (`python3-tk` on many Linux distributions).

The robot bridge runs in a background thread alongside robot code. Copy `ServiceBridge.java` and the adapter for your WPILib year from examples/bridge. It listens only on `127.0.0.1:5810`; access it over an authenticated SSH tunnel. The 2026 adapter requires an explicit maintenance-request flag **and** Driver Station disabled state for every transmission. The 2027 alpha-7 adapter uses `RobotState` and an explicit `CANPort`, compiled separately with JDK 25. Choose a classical-CAN-compatible bus; this MCU cannot transmit CAN FD.

Keep the robot disabled, mechanically secure it and keep outputs inhibited in your robot code throughout maintenance. The bridge rejects or disconnects when permission/disabled state is lost; it cannot prevent someone enabling the robot at the Driver Station. No robot program is stopped or redeployed by this tool. Do not expose this unauthenticated TCP framing directly on a network. The sensor protocol itself has no authentication; CRC detects corruption, not a malicious uploader.

Example using a roboRIO reached over USB; use your controller's actual address/account:

```
ssh -N -L 5810:127.0.0.1:5810 lvuser@172.22.11.2
python host/can_service.py --bridge localhost:5810 discover
python host/service_gui.py
```

Copy the desired 24-hex-digit UID from discovery. Replace `UID` below with it:

```
python host/can_service.py --bridge localhost:5810 blink --uid UID
python host/can_service.py --bridge localhost:5810 configure \
  --uid UID --device 17 --invert 0 --debounce 2
python host/can_service.py --bridge localhost:5810 update \
  --uid UID --file firmware/build/application.sc12
```

Direct SocketCAN uses `--channel can0`. Other commands: `info`, `run`, `rate --milliseconds 10|20|50`. Settings persist after save/reset; status period resets to 10 ms. Existing driver timeout is 100 ms. Traffic is 131/81/51 frames/s at 10/20/50 ms; verify bus jitter physically.

Discovery resolves the complete STM32 96-bit UID independently of assigned CAN IDs, including blank and duplicate-ID devices. Blink confirms physical selection. Before a change, the tool reads back all three UID words. ID conflict checks cover **responding SC12 devices only**; a powered-off board or another Team Use product can still have a conflicting address. Use one maintenance client per CAN network. The service APIs must be reserved against other Team Use devices.

Opening a session suppresses normal telemetry on all participating SC12 boards, and freezes/rebaselines input counts. Sensors remain powered by hardware. Drivers may retain their last sample until its normal freshness timeout; robot code must treat maintenance as unavailable immediately. Close/lease expiry restores normal sampling/telemetry unless Broadcast Disable previously latched it off. Broadcast Disable cancels maintenance and pending transmissions; explicit resume remains necessary for that latch. Never interpret a missing sample as a false input.

Wire protocol

Classical CAN, extended identifiers, eight data bytes, no RTR. Request ID `0x0A08C000`, response ID `0x0A08C040` (Team Use 8, miscellaneous type 10, APIs 0x300/0x301, address 0). Payload: opcode byte, sequence byte, nonzero little-endian session token (16 bits), little-endian value (32 bits). Response opcode OR 0x80 means success; OR 0xC0 means error. Echo sequence/token and validate the expected value. Errors 1-5 mean argument, state/offset, flash, image, and missing boot protection. No reply means an unmatched session/selection or transport failure.

Operation	Value / result
OPEN 0x01	Magic 0x314D4353; broadcast ACKs are identical; five-second lease
PREFIX0/1/2 0x02-04	Three raw UID words; no ACK; each change clears selection
PROBE 0x05	Prefix length 0-96, MSB first within each UID byte; matching nodes reply 1; only 96 selects
CLOSE 0x06	End session, no ACK
INFO 0x10	Index 0 flags/versions, 1 settings, 2 mode version, 3 upload offset, 4 capacity, 5 image validity, 6-8 UID words
BLINK 0x11	Zero; three-second identify LED
SAVE 0x12	ID/invert/debounce/reserved bytes; reserved must be zero; explicit save then reset
BOOT / RUN 0x13/14	Zero; enter bootloader / run verified application
RATE 0x15	10, 20 or 50 ms; volatile status period
SIZE / CRC 0x20/21	Padded application length and CRC32; bootloader only
BEGIN 0x22	Product/layout tag 0x53433134; invalidate/read back manifest before application writes
OFFSET / DATA 0x23/24	Byte offset then four image bytes; ACK next offset after flash readback
FINISH 0x25	Zero; validate complete flash CRC/vectors, commit manifest last, ACK image CRC

INFO index 0 packs service version in bits 0-7, bootloader mode in bit 8, configured in bit 9, boot protection in bit 10, application minor 5 in bits 16-23, and hardware 4 in bits 24-31. Index 1 packs ID/invert/debounce/status-period as bytes; unconfigured ID is 255. Index 2 is major/minor/patch in bits 16/8/0. Index 5 returns 0 or 1.

Host discovery walks binary UID prefixes; matching responders send identical presence frames. Prefix frames are paced; UID readback guards selection before writes. Queries retry four times, probes twice; default pacing is at least 6 ms between requests and a 300 ms ACK timeout. Binary discovery can take seconds per device; intermittent connectivity can omit a node, so repeat discovery before commissioning. Uploads send OFFSET and DATA for each four-byte word, bounding throughput and working around flash stalls. Only the immediately previous exact request is cached; offset/data logic also accepts an identical last word without rewriting it. Gaps, changed duplicate words and stale sessions are rejected.

Image and memory layout

Region	Address / size
Write-protected bootloader	0x08000000 / 8 KiB
Application	0x08002000 / 21 KiB
Image manifest	0x08007400 / 1 KiB
Configuration A/B	0x08007800 / 0x08007C00, 1 KiB each
Reserved SRAM	First 256 bytes: 192-byte vector table, validated recovery cookie

Both linkers reserve at least 1 KiB stack, excluding static data; runtime stack high-water measurement remains pending. The Cortex-M0 application copies vectors to SRAM and remaps them; bootloader handoff clears pending interrupts, stops SysTick/CAN and returns clocks to HSI. Compiled-register tests check vector copying, GPIO/CAN and configuration writes against peripheral models; they do not emulate analog effects or real interrupt timing.

The update file uses `<8s6I`: `SC12UPD1`, format 1, hardware 4, application address, padded length, CRC32, packed version, then image bytes. Length is 192-21504 bytes and a multiple of four. Initial SP must be aligned and within application SRAM, and Thumb reset PC inside application executable bounds. Manifest words are magic 0x49324353, size, CRC, product tag, CRC of the first four words, and final commit 0xC035A19E. Boot verifies manifest, vectors and full application CRC. No update command can write bootloader, configuration or option-byte regions.

Evidence and remaining gates

Tests execute the actual portable C engine with simulated flash, including 2,148 halfword erase/program power-cut positions on a two-page image, retries, corrupt files, wrong sizes, gaps, stale sessions, duplicate IDs and running-version verification. Compiled ARM tests exercise register intent and 34 configuration halfword interruption cases. Real TCP tests cover fragmentation and permission revocation with mock CAN. The Python uploader/GUI, bridge and firmware need end-to-end testing on real hardware before robot use. See qualification and unit programming/test fixture.

16 Programming and unit fixture

Source: docs/FIXTURE.md

Build a reusable fixture around the documented custom TC2050 contacts, ST-Link, a separately current-limited J1 supply, accessible rail test contacts, CAN adapter and three independently driven input fixtures. This document and scripts are ready for the pilot; no physical fixture or unit test has been performed.

Leave voltage caps absent for initial programming. Keep the SWD probe's voltage pin as target reference only. Program/verify the combined image and set boot write protection with program_factory.tcl, then power cycle and verify protection over CAN. Save the programmer log, factory image SHA-256, component lot and full MCU UID. The setup checks STM32F04x device ID 0x445, 32 KiB flash and WRP pages 0-7 (OpenOCD protection blocks 0-1, not 0-7), consistent with the OpenOCD STM32 driver. Qualify the exact probe/OpenOCD version on the first unit.

Configure a unique ID through the service tool. Record internal 5 V, logic 3.3 V and regulated 12 V with a calibrated meter; acceptance windows are 4.8-5.3, 3.15-3.45 and 11.4-12.6 V. These generous unit-screening windows do not replace rail regulation/load/temperature qualification. Power off before fitting selector caps. Exercise each input low and high using the approved bias and compatible test signal; observe fresh protocol-v3 status and count transitions. Verify telemetry after closing maintenance. Power cycle again and verify saved ID/inversion/debounce.

Copy fixture_measurements.template.json, enter measured voltages and set each check true only after actually observing it. Record live identity/protection plus those results:

```
python host/fixture_record.py --channel can0 --uid UID \
  --measurements measured-unit.json --operator NAME --lot LOT \
  --output unit-UID.json
```

Use `--bridge localhost:5810` for a robot tunnel. An incomplete/failed record exits nonzero. Existing output files are never overwritten. The JSON stores operator/lot/time/UID/settings and a hash of the supplied measurements; keep it with the programming log. It does not independently measure voltages or validate operator assertions.

Design qualification still needs 6/12/22 V input tests, both rails at load, shorts, temperatures, representative sensors/cables, vibration, actual flash power cuts, bus-off/controller disconnect and boot/interrupt timing. See VALIDATION.md. Complete the pilot before calling units production-ready or quoting a qualified manufacturing yield.